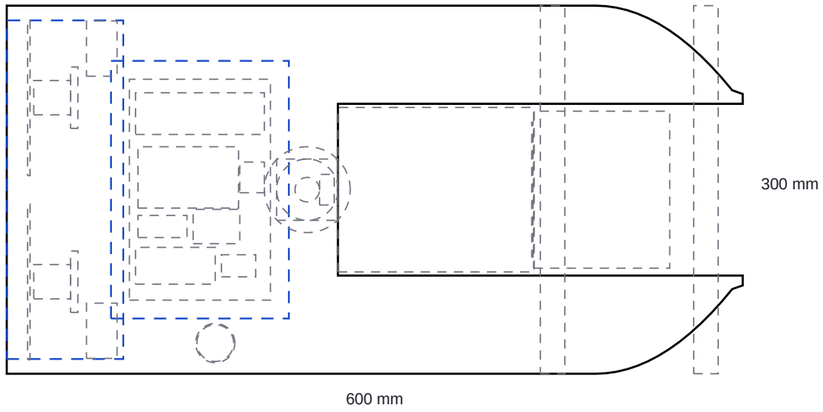
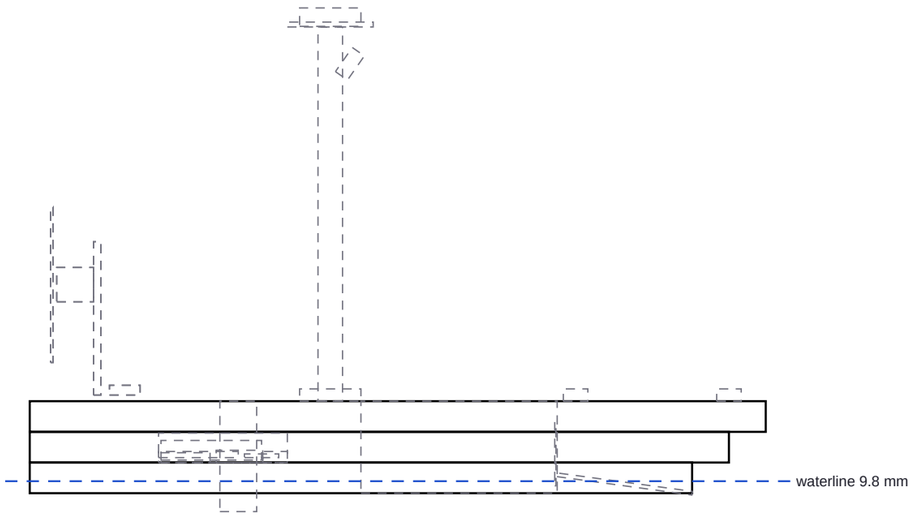
VARUNA U-Hull Block

600 x 300 mm thermocol boat | Easiest to build

PLAN (from above), scale 1:5



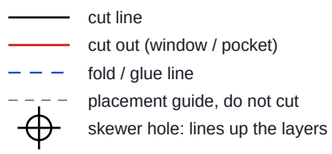
SIDE, scale 1:5

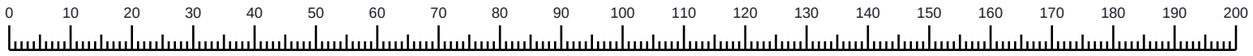


Size	600 x 300 mm, hull 75 mm tall, 396 mm to GPS top
Weight	1144 g empty, 1451 g with 307 g of plastic
Draft	9.8 mm empty, 12.4 mm full
Reserve buoyancy	6.7 kg before the deck goes under
Stability (GM)	833 mm: very stable
Trim	bow 6.2 mm higher than stern empty; slide the battery to level
Collection	1.54 L basket
Camera	340 mm above water, tilted 35 deg down
Battery	LiPo 3S 2200 mAh (24 Wh), about 0.8 h
Thermocol	3 sheets 1000 x 500, or 1 sheet 2 x 3 ft (610 x 915)

How to use these templates

1. Print at 100% / Actual size. Check the 50 mm bar with a ruler.
2. Join tiles by matching the grey 50 mm grid and the tile letters.
3. Stick the paper on thermocol with a glue stick.
4. Cut with a hot-wire cutter or a new snap-off blade, held upright.
5. Stack layers on skewers through the + holes. Glue with Fevicol SH.
6. Never use Fevikwik, spray paint or solvent glue: they melt thermocol.
7. Sand the bow steps smooth, coat with PVA + tissue or packing tape, then water-based acrylic paint.
8. Float-test in a tub before fitting the electronics.





ruler in mm: the 0 to 200 marks must be exactly 200 mm apart

100 mm x 100 mm

How to print on A3 so every part comes out real size

1. Paper: A3, landscape. Printer: 'A3' selected as the paper size.
2. Scaling: 'Actual size' or '100%'. Turn OFF 'Fit to page', 'Shrink oversized pages' and 'Fit to printable area'. On a phone, choose 'Original size' / 'Do not scale'.
3. Print this page only. Measure the square with a steel ruler: both sides must be 100 mm, and the ruler marks must match your ruler. 1 mm off is fine; more is not.
4. If it is wrong, fix the printer setting and print this page again. Do not go on.
5. When it is right, print all the other pages with the SAME settings.
6. Big parts are split across pages (tiles A1, A2, B1...). Cut each tile on its border, lay the next tile over the grey overlap strip, line up the grey 50 mm grid and the measurement ruler, and tape it. Check the long measurement with a tape before cutting.
7. Glue-stick the paper on the thermocol, cut along the solid black line, cut out the red parts, and push a skewer through every + mark.

Line types:

- cut line
- cut out (window / pocket)
- fold / glue line
- placement guide, do not cut
- skewer hole: lines up the layers
- measurement (mm), do not cut

A1

600 overall (to bow tips)

HULL layer 1 (bottom)

overlap: tile A2 goes on top

0 50 100 150 200 250

300 wide

probe hole 32 mm

overlap: tile B1 goes on top

<- must measure 50 mm

A2

ramp wedge

SLOT (cut away) - BOW >

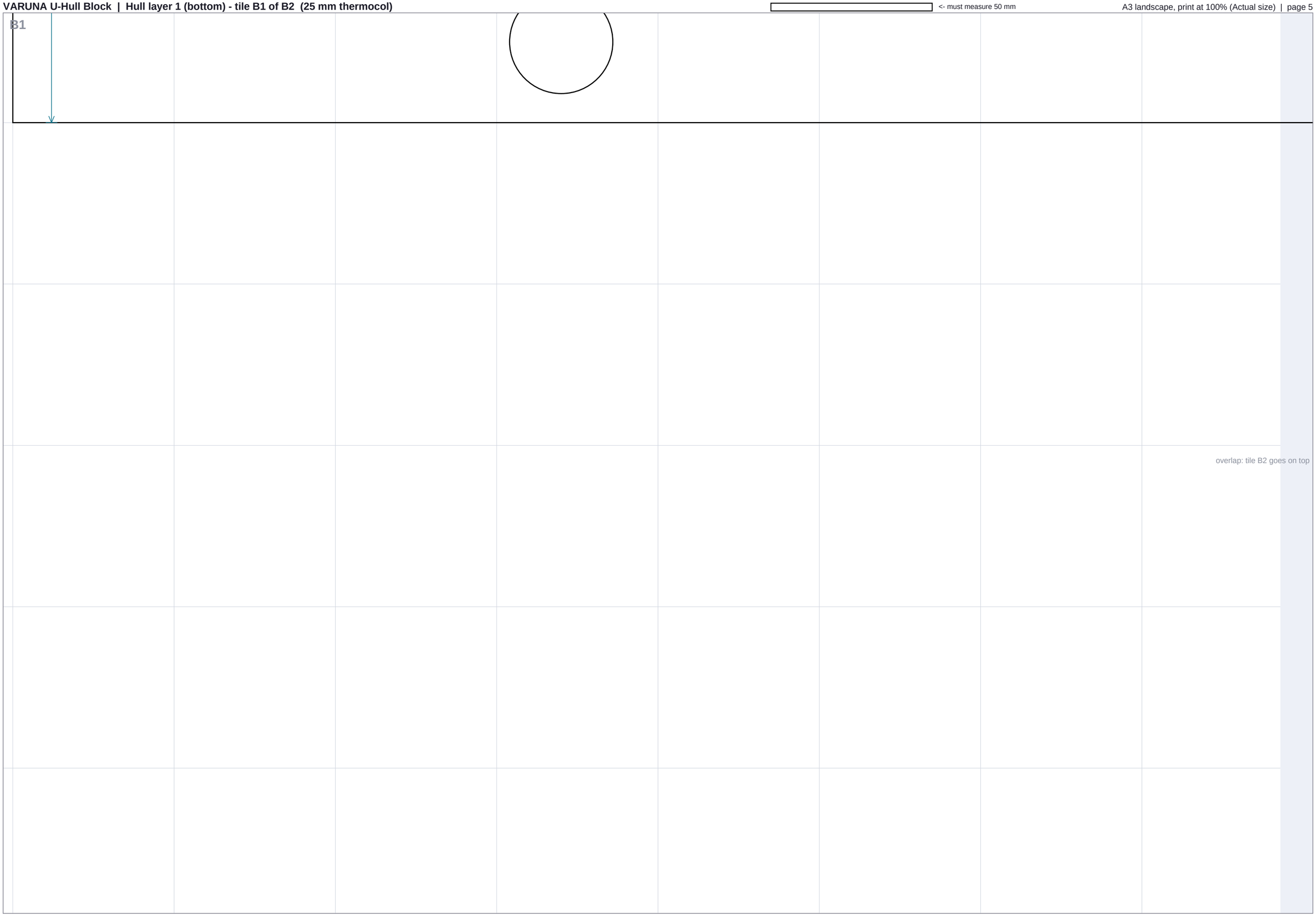
140 slot

330 slot depth

overlap: tile B2 goes on top

<- must measure 50 mm

B1



overlap: tile B2 goes on top

B2

<- must measure 50 mm

A1

600 overall (to bow tips)

HULL layer 2 (middle)

overlap: tile A2 goes on top

0 50 100 150 200 250

300 wide

electronics pocket: cut out

probe hole 32 mm

overlap: tile B1 goes on top

<- must measure 50 mm

A2

SLOT (cut away) - BOW >

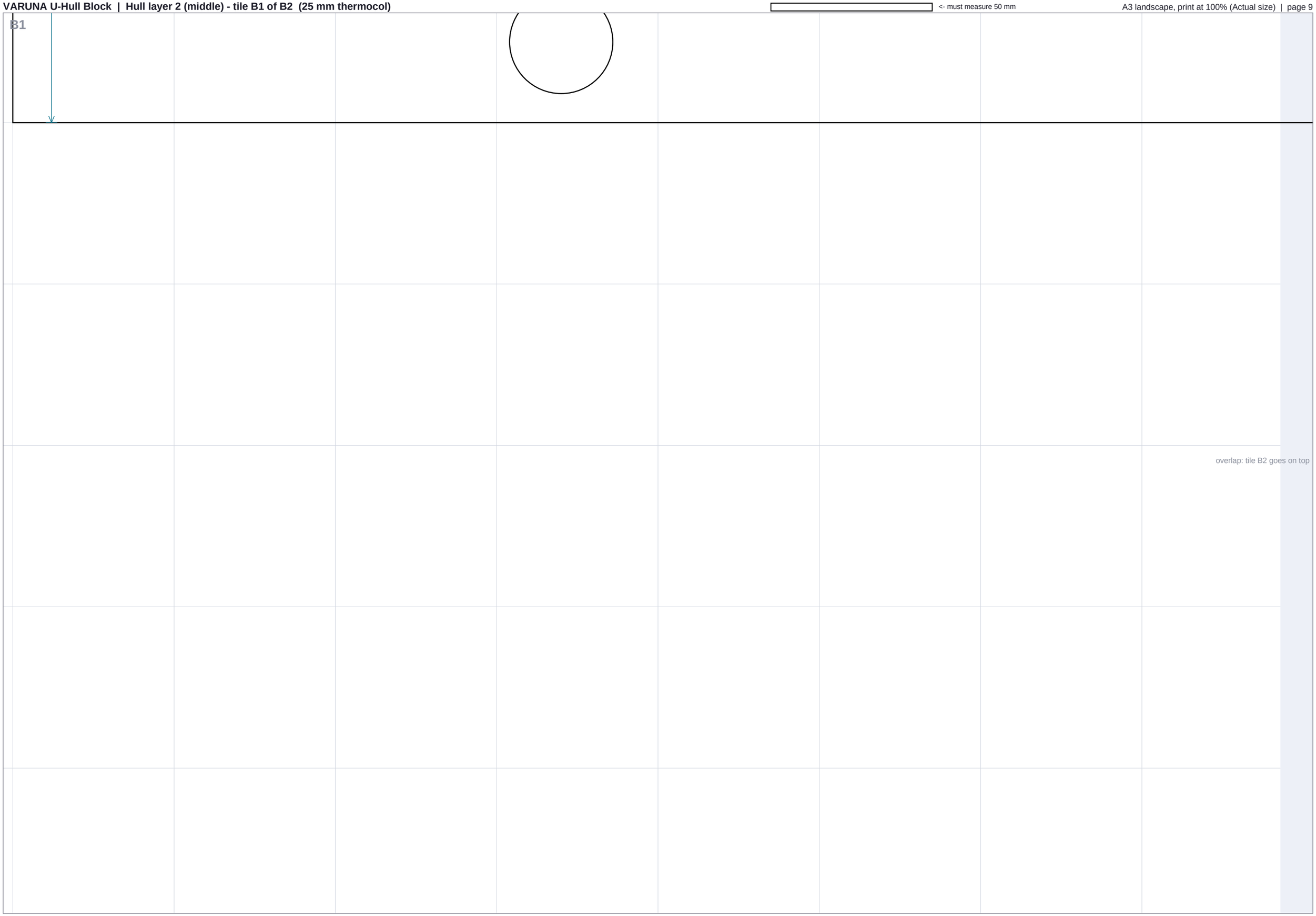
140 slot

330 slot depth

overlap: tile B2 goes on top

<- must measure 50 mm

B1



overlap: tile B2 goes on top

B2

<- must measure 50 mm

A1

600 overall (to bow tips)

basket lip rests here

HULL layer 3 (top)

overlap: tile A2 goes on top

0 50 100 150 200 250

300 wide

electronics pocket: cut out

cable slot

probe hole 32 mm

overlap: tile B1 goes on top

<- must measure 50 mm

A2

SLOT (cut away) - BOW >

140 slot

330 slot depth

overlap: tile B2 goes on top

<- must measure 50 mm

B1

overlap: tile B2 goes on top

<- must measure 50 mm

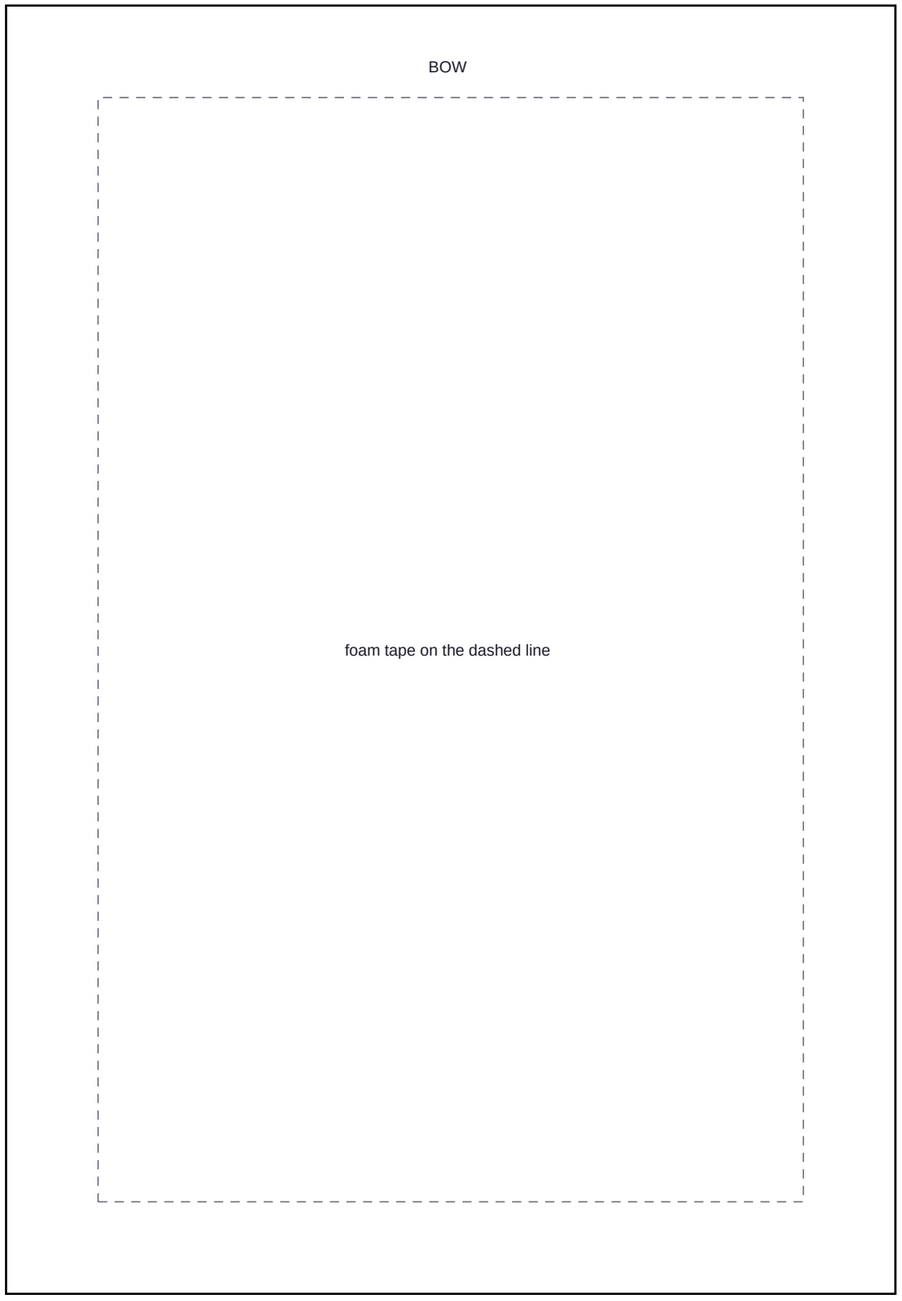
B2

VARUNA U-Hull Block | small parts 1 of 5

<- must measure 50 mm

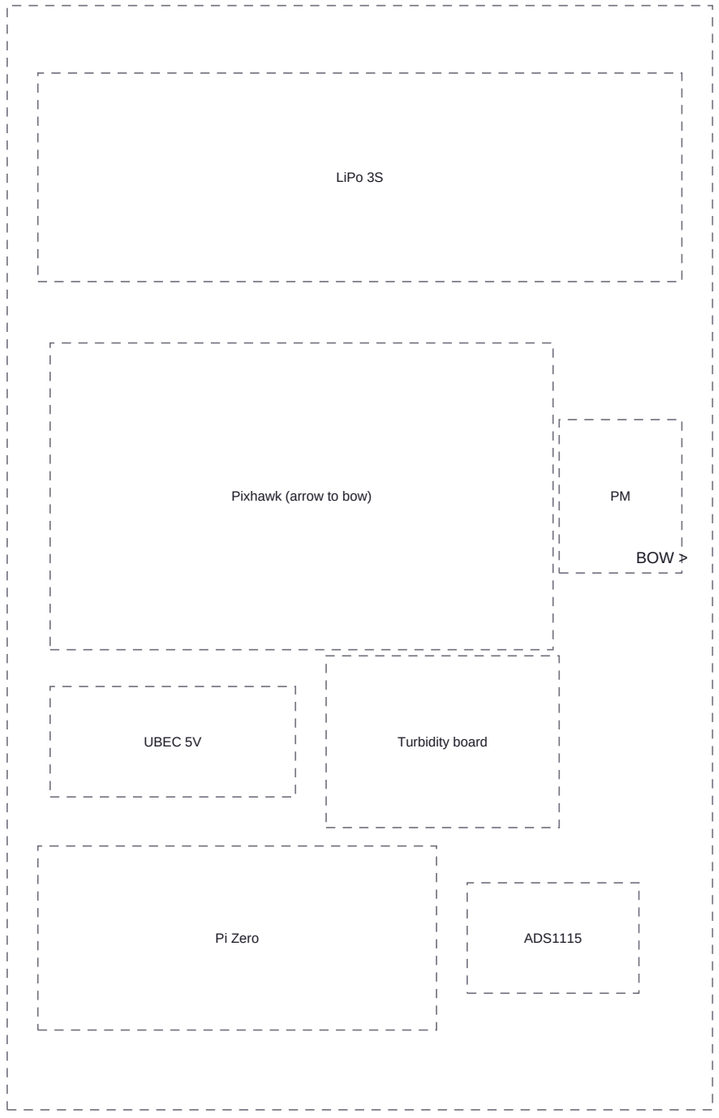
Hatch lid

5 mm sunboard | 145 x 210 mm



Pocket floor guide

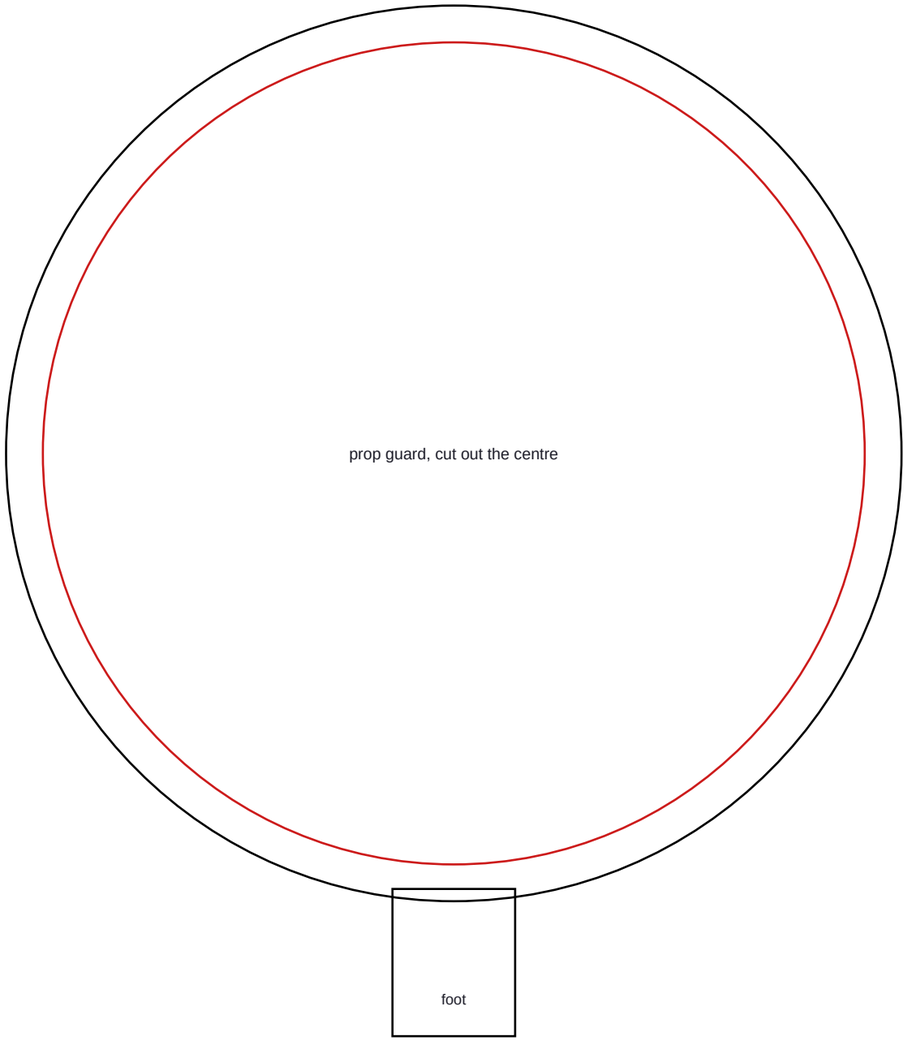
paper guide, glue inside the floor | 115 x 180 mm



<- must measure 50 mm

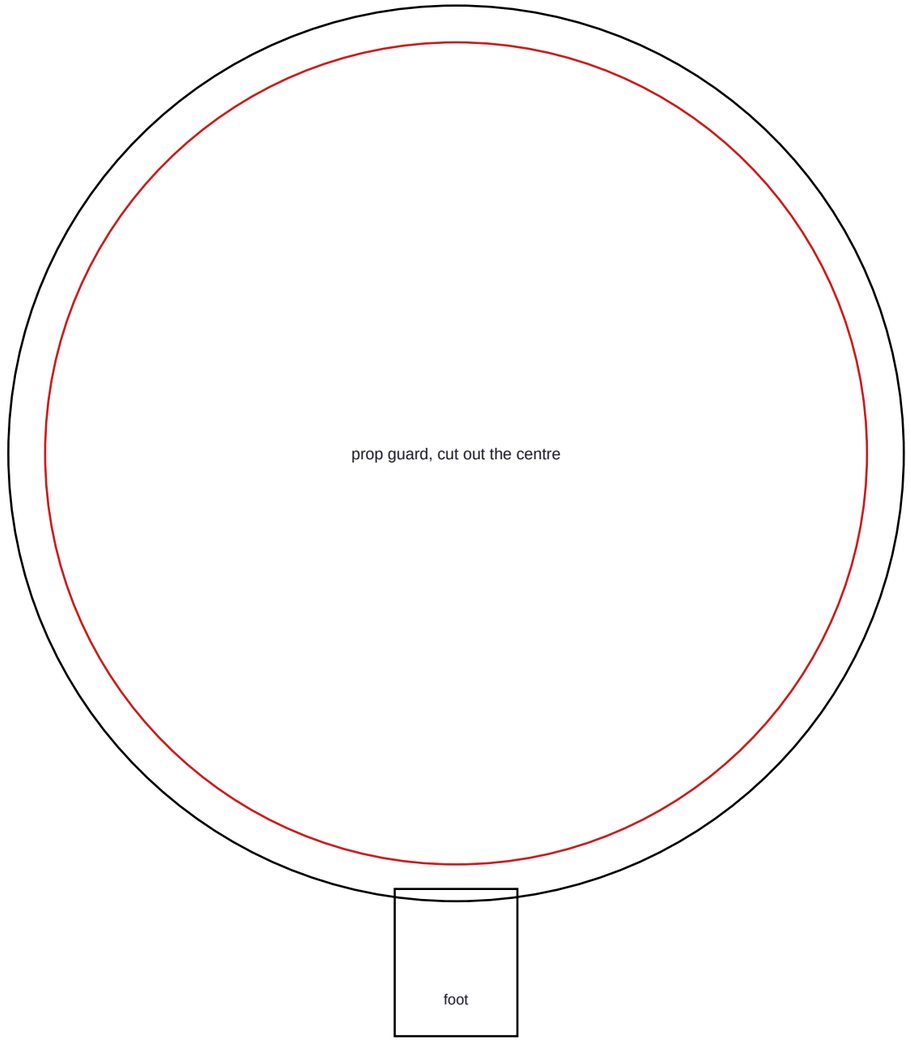
Prop guard ring (left)

5 mm sunboard | 146 x 168 mm



Prop guard ring (right)

5 mm sunboard | 146 x 168 mm

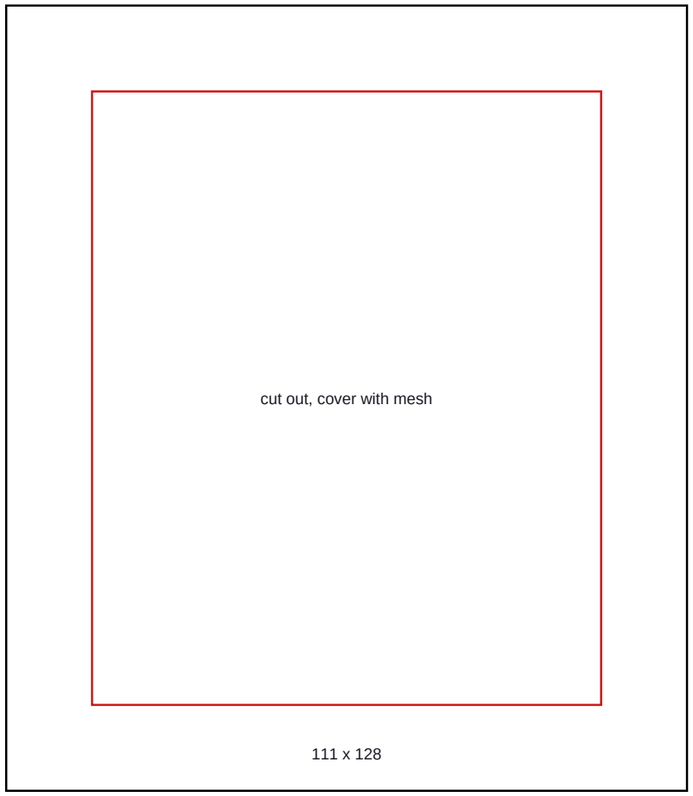


VARUNA U-Hull Block | small parts 3 of 5

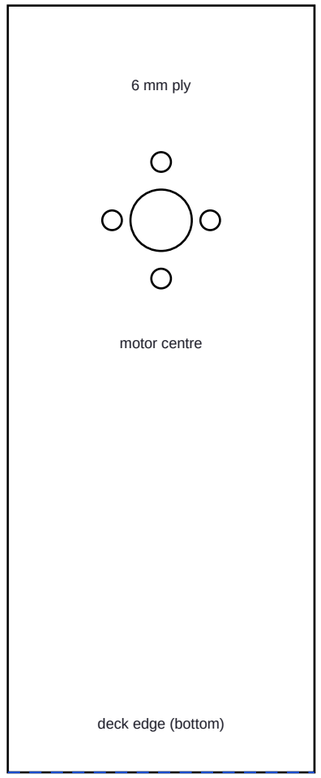
<- must measure 50 mm

Basket bottom
3 mm sunboard | 160 x 128 mm

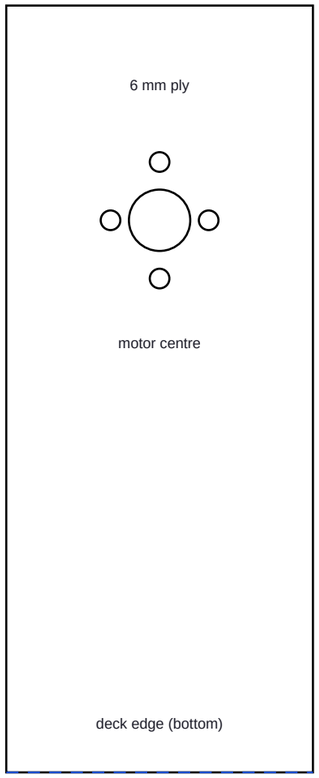
Intake ramp frame
3 mm sunboard | 111 x 128 mm



Motor tower plate (left)
6 mm plywood (or 2 x 5 mm sunboard) | 50 x 125 mm

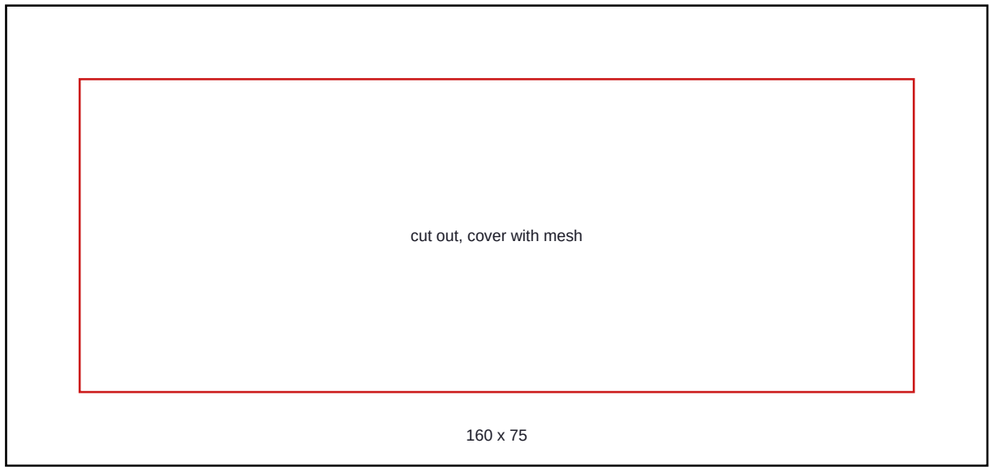


Motor tower plate (right)
6 mm plywood (or 2 x 5 mm sunboard) | 50 x 125 mm

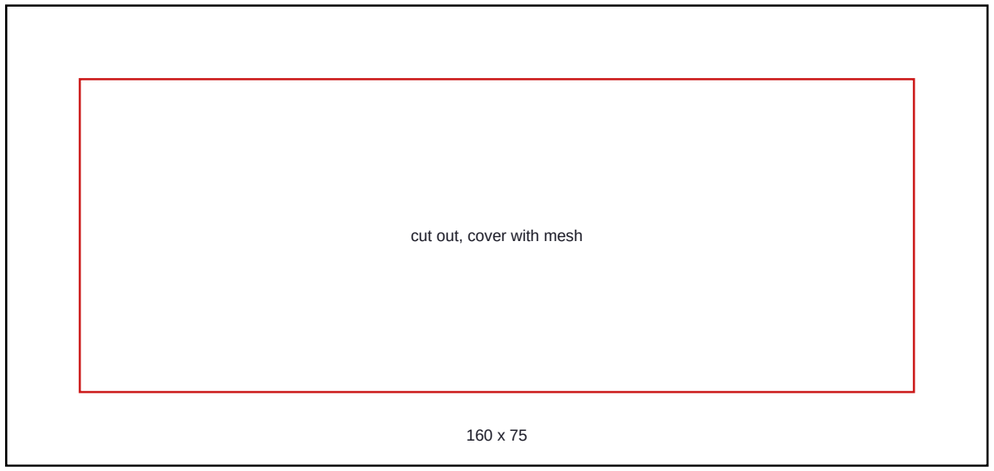


Stern plate
5 mm sunboard | 276 x 95 mm

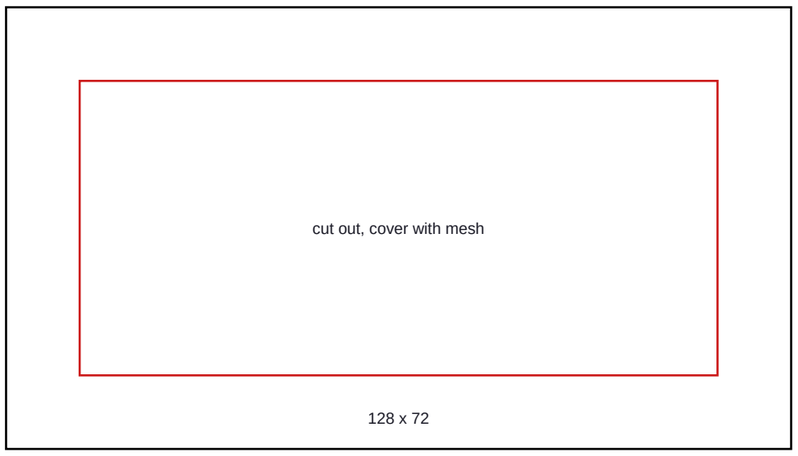
Basket side (left)
3 mm sunboard | 160 x 75 mm



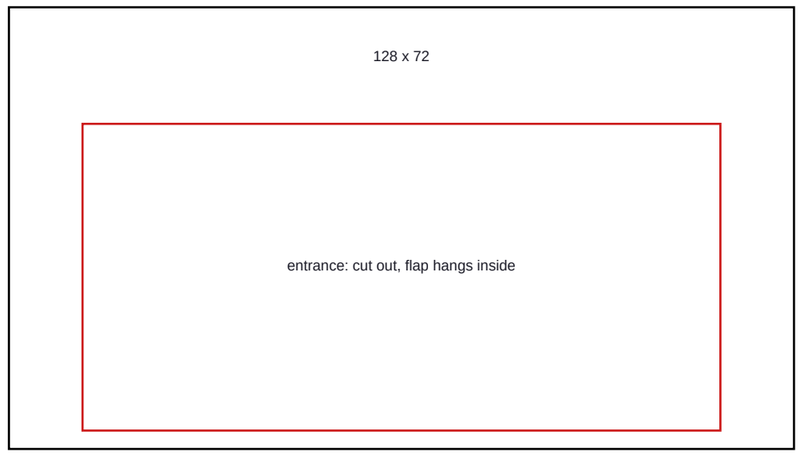
Basket side (right)
3 mm sunboard | 160 x 75 mm



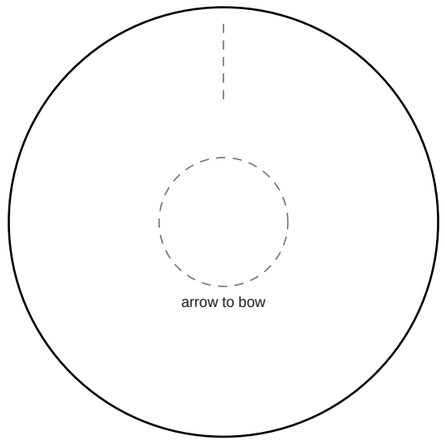
Basket back end
3 mm sunboard | 128 x 72 mm



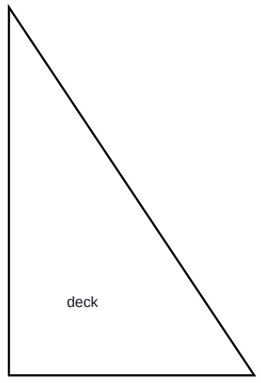
Basket front end (entrance)
3 mm sunboard | 128 x 72 mm



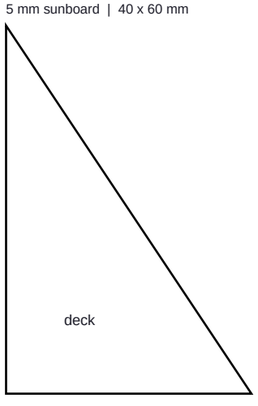
GPS disc
5 mm sunboard | 70 x 70 mm



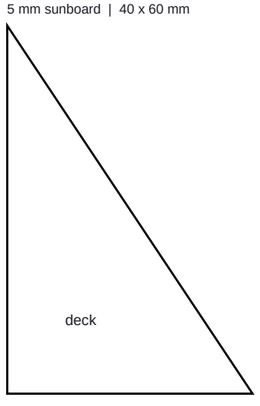
Tower gusset
5 mm sunboard | 40 x 60 mm



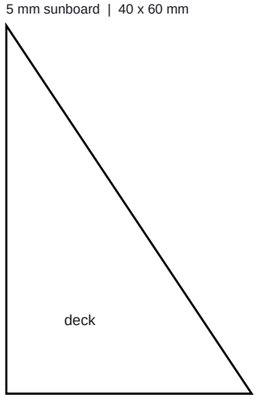
Tower gusset
5 mm sunboard | 40 x 60 mm



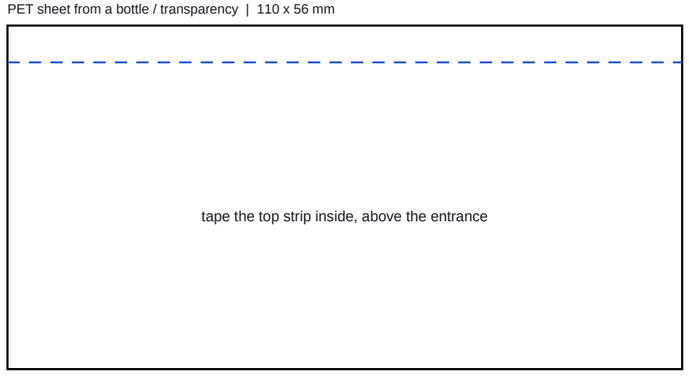
Tower gusset
5 mm sunboard | 40 x 60 mm



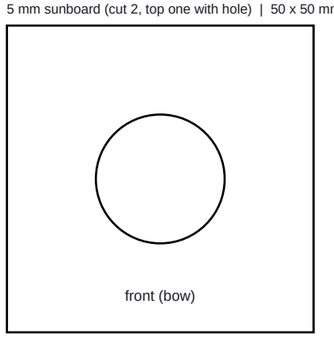
Tower gusset
5 mm sunboard | 40 x 60 mm



One-way flap
PET sheet from a bottle / transparency | 110 x 56 mm



Mast socket layer
5 mm sunboard (cut 2, top one with hole) | 50 x 50 mm



Mast socket layer
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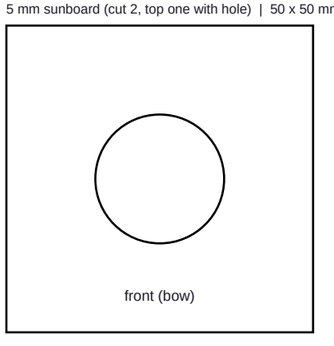
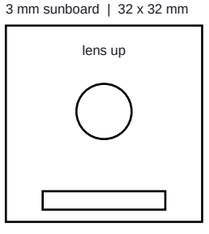


Camera plate
3 mm sunboard | 32 x 32 mm



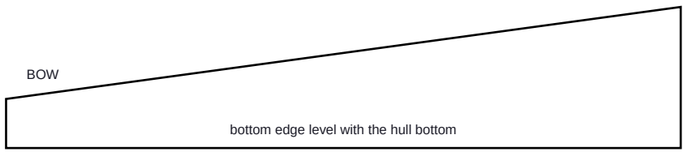
VARUNA U-Hull Block | small parts 5 of 5

<- must measure 50 mm

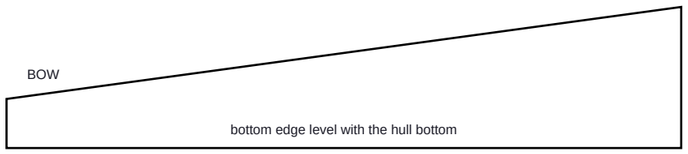
Basket lip (left)
3 mm sunboard | 160 x 28 mm

Basket lip (right)
3 mm sunboard | 160 x 28 mm

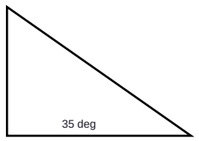
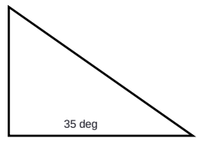
Ramp support wedge
5 mm sunboard (cut 2) | 110 x 23 mm



Ramp support wedge
5 mm sunboard (cut 2) | 110 x 23 mm



Camera wedge side (35 deg)
3 mm sunboard | 30 x 21 mm



Camera wedge side (35 deg)
3 mm sunboard | 30 x 21 mm